

California Tax-Exempt Income — Dual-Axis Credit & Tech/AI-Insulation

Methodology

Sample tearsheet · v2 multi-channel framework · Signal OS muni_credit · 2026-06-07 · **illustrative; not an offer or investment advice**

98/100 avg AI-insulation score	85/100 avg creditworthiness (verified)	4.44% avg yield-to-worst	4.65% avg yield-to-maturity	8.92% taxable-equiv (×2.01)	10.1y duration-to-worst	11 holdings (research universe)
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Executive summary

The problem. A high-bracket Californian — especially in tech — is already heavily exposed to the tech/AI cycle through salary, equity comp (RSUs/options), and portfolio. Yet the obvious tax shelter for their bond allocation, California munis, is at the state level the *most tech-levered government revenue base in the country* (capital-gains / stock-option income tax). Buy a generic CA muni fund and you may be **doubling down on the very risk you're trying to diversify away from**.

The product. A California tax-exempt income strategy selected on **two independent, auditable screens**: (1) **creditworthiness**, verified bond-by-bond against primary sources, and (2) **tech/AI income-channel insulation**, measuring exposure across **nine** distinct transmission channels at the bond-security level. The result: double-tax-exempt income (~4.4% YTW / 8.9% taxable-equivalent) from credits whose *creditworthiness does not deteriorate when tech/AI wealth evaporates*.

Why it's different — and defensible. No generic CA muni ETF offers **income-channel diversification**. Every score is transparent and every exclusion is auditable (see worked examples). And we are deliberately precise about what this is *not*: not a price hedge, not alpha, not a market forecast — just tax-exempt income, verified credit quality, and a diversification axis no one else screens for.

The pitch — two independent screens

Creditworthiness screen (verified diligence). Each bond's pledged security is verified against primary sources and the full trail is retained: the claim (the security/lien), how it was checked, the authority that confirms it (EMMA official statements, CA Gov Code, CDIAC default-draw, rating actions), and the verdict (Verified / Verified-wrap / **Unverifiable**). The diligence is auditable, not a black box — and *Unverifiable ≠ clean*: a bond whose security we have not actually read is excluded, not assumed good.

Tech/AI insulation screen (multi-channel). A tech/AI bust reaches California muni credit through nine distinct channels, not one. We score each separately and at the bond-security level (the pledged revenue), not the issuer's general economy — so a property-tax GO levy scores insulated even if the issuer's *budget* feels pension or sales-tax pressure. The composite score is insulation across all nine channels.

Tech/AI Transmission Channels

An AI/tech bust reaches CA muni credit through **nine distinct channels**, not one. We score each separately and at the **bond-security level** (does the channel reach the *pledged revenue*?). “Wt” = weight in the insulation composite; “Port. exp” = this portfolio's average exposure to that channel.

Channel	What it is — how a tech/AI bust reaches the bond	Wt	Port. exp
State-GF capital-gains income tax	CA's General Fund leans on capital-gains / stock-option / IPO income tax — the most tech-levered state revenue base in the U.S. A tech/AI bust evaporates it. → State GO & appropriation debt .	0.30	0%
Tech-hub property / assessed value	Tech wealth bids up Bay-Area/Silicon-Valley property; a bust softens assessed value (Prop-8 reductions, slower turnover). → tax-increment (TAB) & land-secured bonds .	0.15	3%
CalPERS/CalSTRS pension contributions	CalPERS/CalSTRS hold large equity + growing VC/tech; an equity crash → investment losses → higher required contributions for every CA local government. → operating budgets / appropriation debt (NOT protected GO debt service).	0.12	0%
Tech-firm taxpayer/employer concentration	Some local credits lean on one tech firm as a dominant taxpayer/employer (Apple-Cupertino, Google-Mtn View). A firm contraction hits them directly. → concentrated tech-hub local credits .	0.10	0%
Hospital balance-sheet (investments/philanthropy)	Silicon-Valley hospital reserves (investment income), tech-donor philanthropy, and elective volumes (tech employment/insurance) are tech-linked. → CHFFA hospital conduits .	0.10	4%
Commercial real estate / office	Tech layoffs + AI-driven lower office demand → SF/SV office vacancy ('doom loop') → falling commercial AV & assessments. → CRE-heavy city GF, TABs, assessment districts .	0.08	0%
Bay-Area housing prices	Tech wealth drives Bay-Area home prices; a bust → mortgage delinquency. → CalHFA single-family + multifamily / workforce housing .	0.06	10%
Local sales / hotel (TOT) tax	Tech-worker spending + tech business travel/conferences feed local sales tax + hotel (TOT) tax. → tech-hub city general funds .	0.05	0%
Transit / airport	Bay-Area transit ridership + business travel (SFO, BART, bridges) ride tech employment/travel (remote-work/AI compounding). → transportation revenue credits .	0.04	0%

The portfolio is built from pledges (property-tax GO debt service, essential-service revenue, insured/agency) structurally disconnected from **all nine** channels — not just income tax. **That** is the defensible meaning of “insulated.” Note the budget-vs-security distinction: the pension channel can squeeze a school district's operating budget while its GO debt-service levy stays untouched — we score the pledge, not the issuer's general economy.

Creditworthiness Check (verified diligence)

Every credit claim is verified against a primary authority and **documented end-to-end**: the claim, how it was checked, the authority that confirms it, and the verdict (Verified / Verified-wrap / **Unverifiable**). “CORE” dimensions drive the v2 creditworthiness score; “EXTENDED” are sector-specific diligence layers from the detector library, applied where the pledge warrants.

Dimension	What we verify	Authority / source	In score
Security & lien	What is pledged and how senior your claim is — statutory first lien > revenue covenant > naked conduit > appropriation/lease. The single most important credit fact.	EMMA official statement + CA Gov Code	CORE (base score)
External rating	Independent agency rating (S&P / Moody's / Fitch) as third-party corroboration of the credit.	Rating agencies	CORE (overlay)
Default & draw history	Has the obligor ever defaulted or drawn on its debt-service reserve? A recorded event is a hard exclude.	CDIAC DebtWatch default-draw index (live)	CORE
Liquidity & disclosure	Does the bond actually trade (so you can mark & exit), and is the issuer current on disclosure? Stale / never-traded = penalty.	EMMA RTRS trade tape	CORE (penalty)
Covenant compliance	Debt-service coverage, additional-bonds test, reserve maintenance — is the issuer meeting its covenants?	EMMA continuing disclosure / audits	EXTENDED (covenant_breach)
Fiscal distress	Going-concern flags, late filings, county fiscal certifications (school GO).	Auditor opinions / EMMA / COE AB-1200	EXTENDED
Pension / OPEB burden	Pension/OPEB funded ratio and 5-year trajectory — the slow-burn local-budget risk.	CalPERS/CalSTRS valuations	EXTENDED

Guiding rule: **UNVERIFIABLE** ≠ *clean*. A bond whose security we have not actually read is excluded until it is — a strong agency rating does not substitute for the unread lien (see *El Camino*, below).

Worked example — Included

INCLUDED — Duarte Unified School District

AI-insulation 99 · creditworthiness 90 · YTW 4.90%

Channel exposure: Tech-hub property / assessed value 5%

Diligence component	Claim	Source / authority	Finding
security lien	Unlimited ad-valorem tax pledge + SB-222 statutory FIRST LIEN on the levy	CA Government Code §53515 (SB-222); EMMA OS	VERIFIED
cdiac default draw	No recorded CDIAC default/draw for this obligor	CDIAC DebtWatch default-draw index	CLEAN

Decision: passes BOTH gates — VERIFIED SB-222 statutory first lien on the ad-valorem levy (credit 90), and insulated across all nine channels (99).

Worked example — Excluded for creditworthiness

EXCLUDED — creditworthiness — Oroville Public Financing Authority AI-insulation 76 · creditworthiness 18
illustrative, drawn from the live CDIAC index — not a portfolio candidate

Diligence component	Claim	Source / authority	Finding
security lien	Marks-Roos pooled financing-authority revenue (land-secured / assessment pool) — junior, no statutory lien	EMMA / CDIAC	WEAK-STRUCTURE
cdiac default draw	14 recorded DEFAULTS + 6 reserve draws, 1996–2004	CDIAC DebtWatch default-draw index (live)	SEVERE / EXCLUDE

Decision: Excluded on creditworthiness. The CDIAC default-draw record shows a **documented serial-default history** (14 defaults, 6 reserve draws). A recorded default is the hardest credit fail there is — an actual failure to pay, not a thin score — so the **cdiac_default_draw** screen hard-excludes it regardless of yield or insulation. The credit gate at its sharpest.

Worked example – Diligence resolved (was unverifiable)

DILIGENCE RESOLVED — credit **VERIFIED**, now excluded for AI — El Camino Hospital (CHFFA conduit 13032U AI-insulation 88 ·

creditworthiness **88** · YTW 1.21%

Channel exposure: Tech-hub property / assessed value 14%, Tech-firm taxpayer/employer concentration 28%, Hospital balance-sheet (investments/philanthropy) 70%

Diligence component	Claim	Source / authority	Finding
security lien	CHFFA conduit (unenanced): Gross Revenues pledge under a Master Trust Indenture (Obligated	EMMA OS emma.msrb.org/ES1016029-ES795266-ES1196616.pdf (	VERIFIED
external rating	Agency rating AA/Aa3 (El Camino — large SV system, AA)	S&P/Moody's/Fitch	VERIFIED
cdiac default draw	No recorded CDIAC default/draw for this obligor	CDIAC DebtWatch default-draw index	CLEAN

Decision: Previously flagged **UNVERIFIABLE** (covenants unread). We pulled the official statement and verified the security: a **Gross Revenues pledge under a Master Trust Indenture, $\geq 1.15\times$ rate covenant** (1.10 $\times$ default floor), no debt-service reserve — an investment-grade package, corroborated by the AA rating. Credit re-scored **73 $\rightarrow$ 88 (VERIFIED)**. It is now excluded for the *right* reason — **AI-channel exposure** (insulation 88 < 90: hospital balance-sheet + Silicon-Valley concentration), not for an unread document. *The diligence loop: UNVERIFIABLE is a to-do, not a verdict.*

Worked example — Excluded for AI correlation

EXCLUDED — AI correlation — California State General Obligation AI-insulation **66** · creditworthiness **86** · YTW 4.73%

Channel exposure: State-GF capital-gains income tax 100%, CalPERS/CalSTRS pension contributions 30%, Commercial real estate / office 5%, Local sales / hotel (TOT) tax 10%

Diligence component	Claim	Source / authority	Finding
security lien	State GO full faith & credit; constitutional 2nd-priority on the General Fund	CA Const. Art. XVI; CA Treasurer	VERIFIED
external rating	Agency rating AA-/Aa2 (CA State GO current AA-)	S&P/Moody's/Fitch	VERIFIED
cdiac default draw	No recorded CDIAC default/draw for this obligor	CDIAC DebtWatch default-draw index	CLEAN

Decision: strong, VERIFIED credit (86) but fails insulation (66): **100% exposed to the state-GF capital-gains income-tax channel** — the exact tech/AI pipe (CA State GO fell ~6 notches in dot-com). The orthogonality in action: good credit, wrong channel.

Portfolio holdings (11, equal-weight)

Obligor	Security	AI-ins	Credit	YTW %	YTM %	Dur-w
Duarte Unified School District	school go sb222	99	90	4.90	4.90	14.5
Alpine County USD GO (SUB for Alpi	school go sb222	99	90	5.00	5.05	15.9
Downey Unified School District GO	school go sb222	99	90	4.24	4.62	7.9
Sanger Unified School District GO	school go sb222	99	90	5.54	5.54	12.8
Pomona Unified School District GO	school go sb222	99	90	4.77	4.77	14.8
Aldersly (Garden Retirement Commun	ccrc calmortgage	96	84	4.12	4.65	3.6
Clovis Unified School District GO	school go sb222	99	82	3.84	3.88	10.6
Metropolitan Water District of Sou	water	100	80	3.80	4.65	4.2
Marina Coast Water District (Rev)	water	100	80	3.05	3.27	6.9
CalHFA program bond	calhfa	96	80	4.60	4.85	5.9
Bethany Home Society of San Joaqui	ccrc calmortgage	96	76	4.98	4.99	14.6

Stats: avg AI-insulation 98 · avg creditworthiness 85 · YTW 4.44% / TEY 8.92% · duration-to-worst 10.1y / to-maturity 13.9y · 6 school GO (SB-222 lien) / 2 water / 2 CCRC (Cal-Mortgage) / 1 CalHFA.

Selection rationale — why each name is in or out

Included (11) — clears both gates

- **Duarte Unified School District** (ins 99 / cred 90) — Unlimited ad-valorem school GO under an SB-222 statutory FIRST LIEN — the strongest security in the book and structurally disconnected from every tech/AI channel. The core archetype.
- **Alpine County USD GO (SUB for Alpine)** (ins 99 / cred 90) — SB-222 statutory-lien school GO; property-tax-secured, fully tech-insulated, 5.0% yield-to-worst.
- **Downey Unified School District GO** (ins 99 / cred 90) — SB-222 statutory-lien school GO, insulated; premium-callable, so its realized (to-worst) duration is shorter — handy for the intermediate-duration target.
- **Sanger Unified School District GO** (ins 99 / cred 90) — SB-222 statutory-lien school GO; the highest yield in the book (5.5% YTW) with full insulation.
- **Pomona Unified School District GO** (ins 99 / cred 90) — SB-222 statutory-lien school GO, insulated, 4.8% YTW.
- **Aldersly (Garden Retirement Communit** (ins 96 / cred 84) — Senior-living revenue wrapped by Cal-Mortgage/HCAI — a state-program credit substitution (~AA-) carrying the credit, over a pledge orthogonal to the tech/AI channels.
- **Clovis Unified School District GO** (ins 99 / cred 82) — SB-222 statutory-lien school GO, insulated; credit a notch lower (82) only because the trade mark is stale — a disclosure flag, not a security weakness.
- **CalHFA program bond** (ins 96 / cred 80) — State housing-finance program bonds — over-collateralized mortgage pools; only a modest housing-price channel, comfortably inside the insulation gate.
- **Metropolitan Water District of South** (ins 100 / cred 80) — Metropolitan Water District essential-service revenue — usage-based cash flow orthogonal to all nine channels; about the most recession- and tech-proof pledge there is.
- **Marina Coast Water District (Rev)** (ins 100 / cred 80) — Essential-service water revenue, insulated; shorter duration.
- **Bethany Home Society of San Joaquin** (ins 96 / cred 76) — Cal-Mortgage-insured senior living; the state wrap does the credit work; insulated.

Excluded (9)

- **El Camino Hospital (CHFFA conduit 13)** (ins 88 / cred 88) — Now VERIFIED investment-grade (gross-revenue pledge, $\geq 1.15\times$ rate covenant, AA) — but excluded for the hospital-balance-sheet channel (Silicon-Valley investment income, philanthropy, elective volumes): insulation 88. Good credit, wrong place.
- **Stanford Health Care (CHFFA conduit)** (ins 88 / cred 88) — VERIFIED AA- ($\geq 1.25\times$ rate covenant); the same hospital + Silicon-Valley-concentration exposure pulls insulation to 88. Excluded for AI, not credit.
- **California State General Obligation** (ins 66 / cred 86) — The orthogonality case in one bond: VERIFIED AA credit (86) but 100% exposed to the state-GF capital-gains income-tax channel — the exact pipe a tech/AI bust runs through (it fell ~6 notches in dot-com). Insulation 66.
- **Anaheim Housing & Public Improvement** (ins 96 / cred 72) — Insulated (96) but excluded on CREDIT — a municipal financing-authority lease/revenue pledge scores below the 75 floor; softer security than a statutory-lien GO.
- **Anaheim Housing & Public Improvement** (ins 80 / cred 70) — Tax-increment (RPTTF) bond — its security IS assessed-value growth, the property/AV channel a tech bust hits; insulation 80, below the gate.
- **Successor Agency to the Fontana Rede** (ins 80 / cred 70) — Tax-increment bond — property/AV channel exposure, insulation 80.
- **Cloverdale CDA Successor Agency (TAB)** (ins 80 / cred 70) — Tax-increment bond — property/AV channel exposure, insulation 80.
- **County of San Diego COP (SUB for Pen)** (ins 86 / cred 64) — Fails BOTH — a pension-obligation certificate of participation (annual-appropriation + abatement risk, no statutory lien → credit 64) with pension/appropriation channel exposure (insulation 86).
- **Successor Agency to the San Jose Red** (ins 74 / cred 62) — The most tech-exposed name here — Silicon-Valley tax increment (property/AV + CRE + tech-firm-concentration, all SV-boosted → insulation 74) plus a never-traded mark dragging credit to 62. Fails BOTH gates.

What this IS — and is NOT

IS

- Tax-exempt (fed+CA) income, primary-source credit-vetted
- **Income-channel diversification** for tech-comp-concentrated CA holders — credit insulated from the tech/AI cycle across nine channels
- Two transparent, auditable, orthogonal scores

IS NOT

- A price hedge / return anti-correlation (that's a duration bet that **inverted -30% in 2022**)
- Alpha — not return-backtested vs a CA muni ETF
- A *validated predictive* rating — scores are STRUCTURAL classifications + diligence verdicts for selection, not forecasts

Honest limits. (1) **11 holdings, not 20** — the dual-axis gate clears only 11 of the 20-name *research* universe; a diversified 20 needs a broader CA screen (same archetypes, more names). (2) **Long duration** (10.1y to-worst) — production targets **intermediate ~4-7y** via maturity selection; this is income + credit quality, NOT crash protection. (3) Scores are structural, not validated betas. (4) Single-state (CA) concentration; some sectors trade thinly.

Reproducible: channel_scoring_v2.py, build_v2_portfolio.py, data/channel_scores_v2.json, data/v2_portfolio.json. Sources: EMMA RTRS (yields/terms), CA Gov Code (SB-222), CDIAC DebtWatch (default-draw), HCAI/Cal-Mortgage, agency ratings. Yields gross of fees, as-of 2026-06-05/07. Non-claims reflect two adversarial reviews and are stated deliberately. Illustrative; not an offer, solicitation, or investment advice.